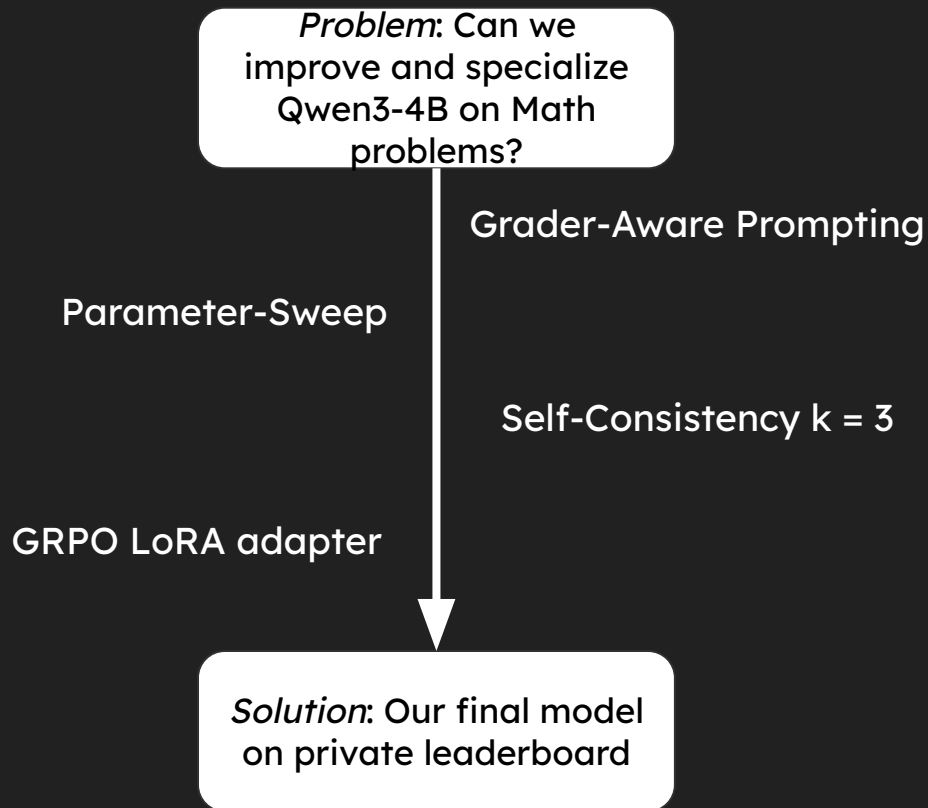


CSE 151B - Project Results

By: Ketan Mittal, Anirudh Annabathula, and Kaii Bijlani

Team Name: *Team Name*

Summary



Attempts:

- Higher Context Limits
- Higher self-consistency parameters
- Running different Quantitizations

Private Leaderboard
Runs

0.515 -> .699

Introduction

Team Introduction



Ketan Mittal

- Grader Analysis
- Prompt Design
- Sampling-Parameter Sweep
- Final private run
- Bug-fixes
- Submission Pipeline



Anirudh Annabathula

- Inference Instructure
- GRPO Training Setup with GPUs
- Presentation



Kaii Bijlani

- Dataset Analysis
- Claude answer filter for dataset
- Presentation

Methodology

Data Processing

Dataset:

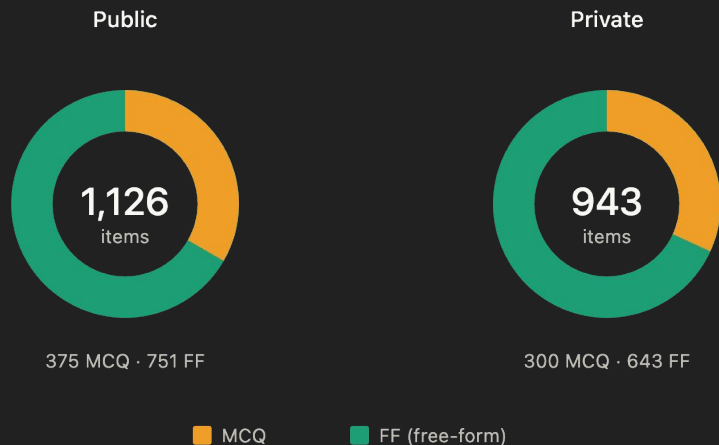
- 1126 Public + 943 private examples
- Multiple Choice (MCQ) + Free Response (FF)
- 8 Different math domains (algebra -> discrete)

Data Processing:

- Claude verified and corrected the answers for the public dataset (1113/1126 were correct)
- 44-item eval slice

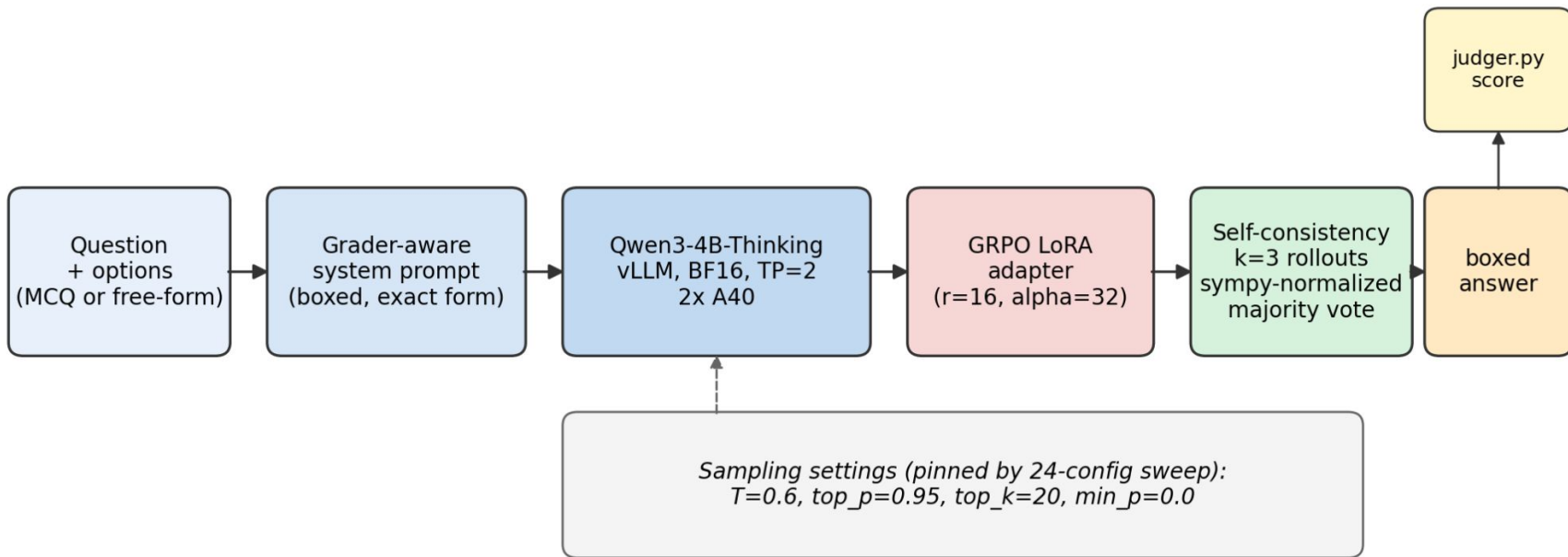
Grader:

- Tolerance of 10^{-8}
- Answers must be either a list or singular symbol or number

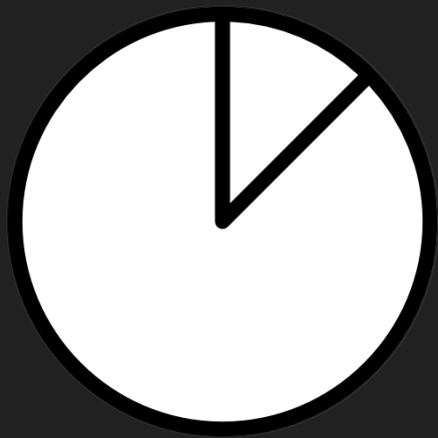


Deep Learning Model

Inference pipeline (sampling sweep + GRPO LoRA)



Engineering Tricks



Ran it on small eval chunks to get a good understanding of the model performance with a change.

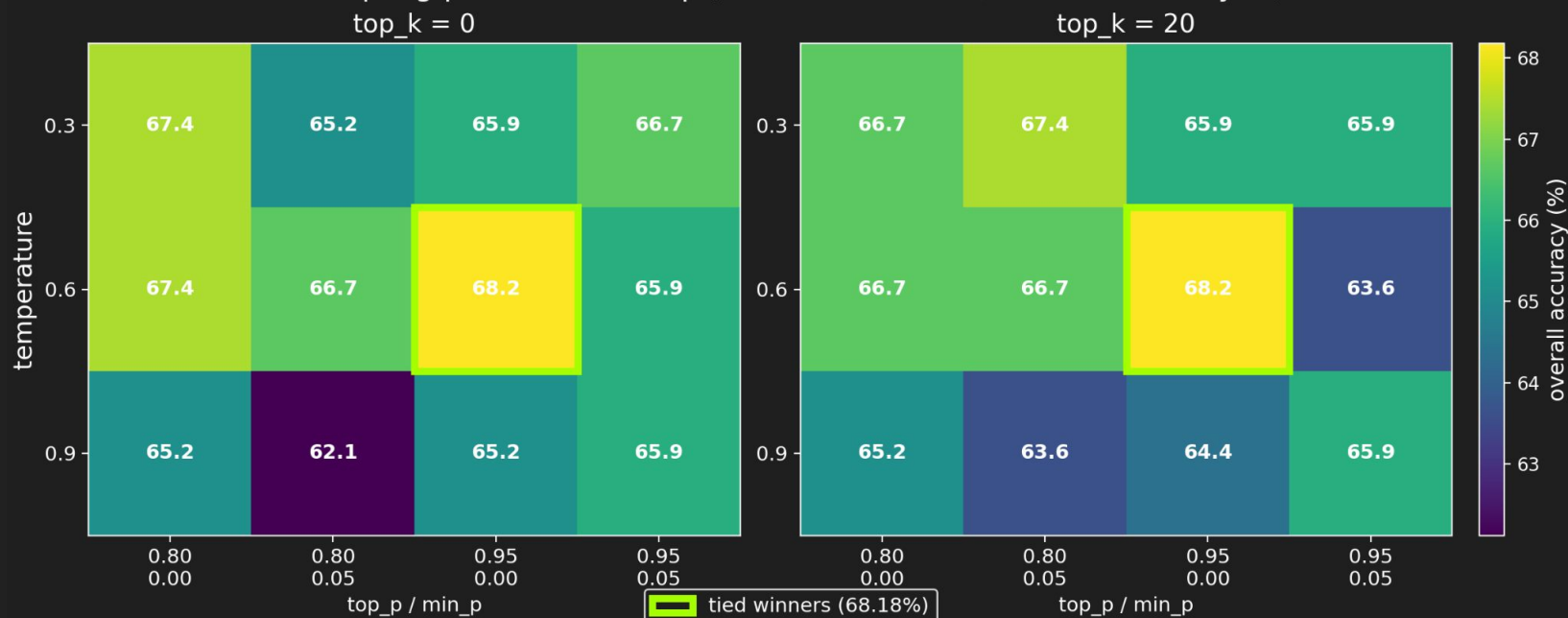


Used bash scripts to run the notebook in order to run tasks in the background when doing other work.

Experiments

Experiment 1

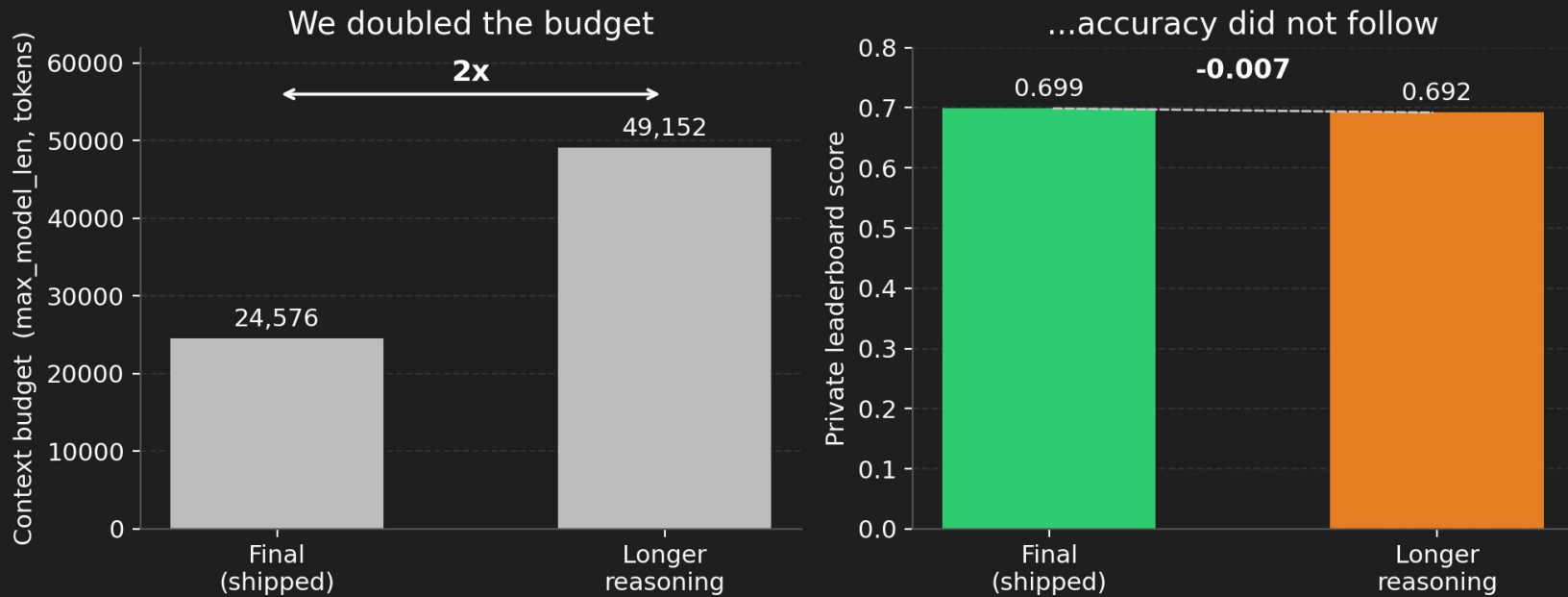
Sampling-parameter sweep (44-item eval slice, overall accuracy %)



- Winner (top_k=0) ties Qwen-recommended (top_k=20) at 68.18% -> pinned recommended config downstream.
- min_p=0.05 consistently hurts free-form; top_k barely matters.

Experiment 2

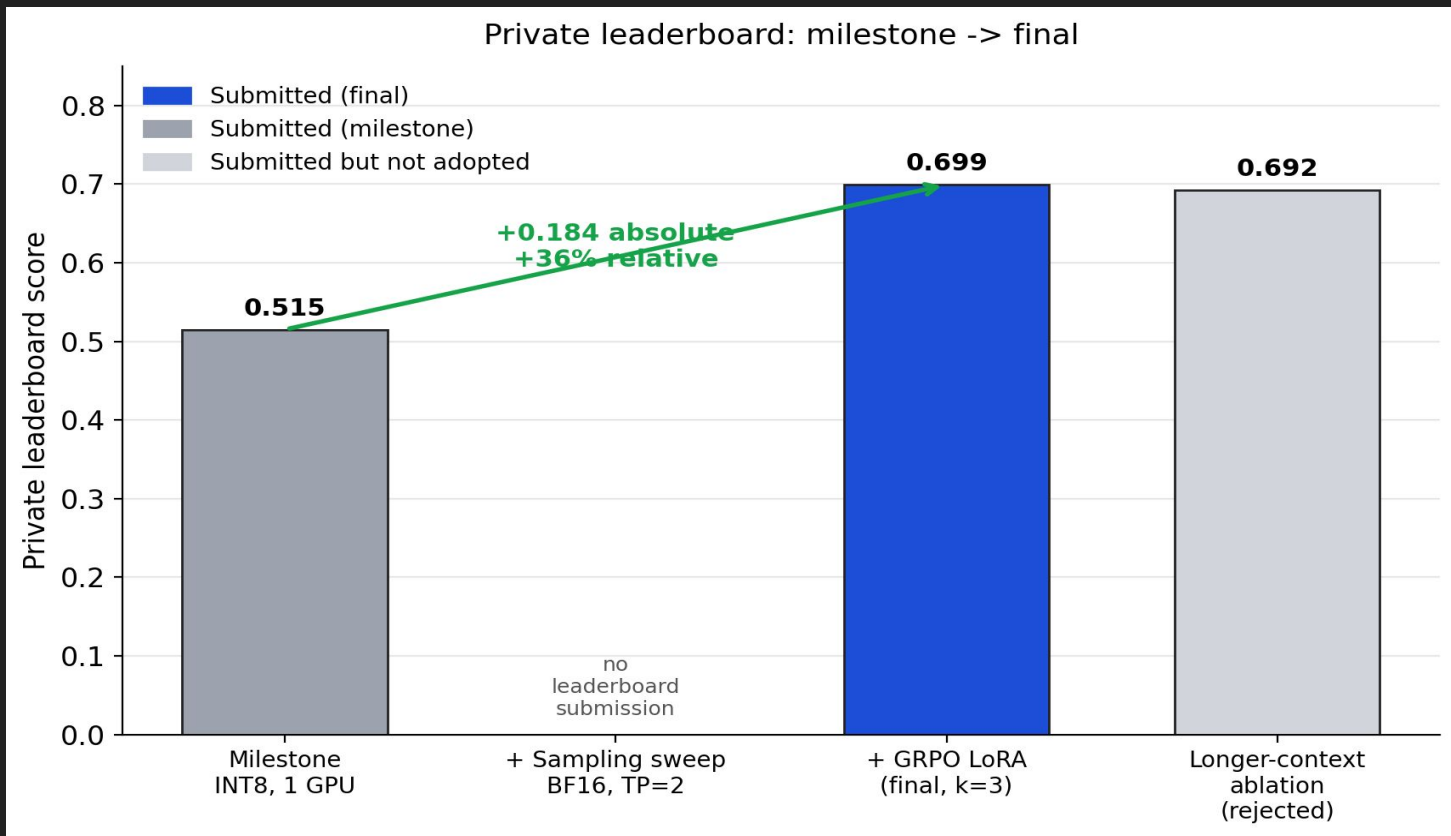
More room to reason did not help (-0.007)



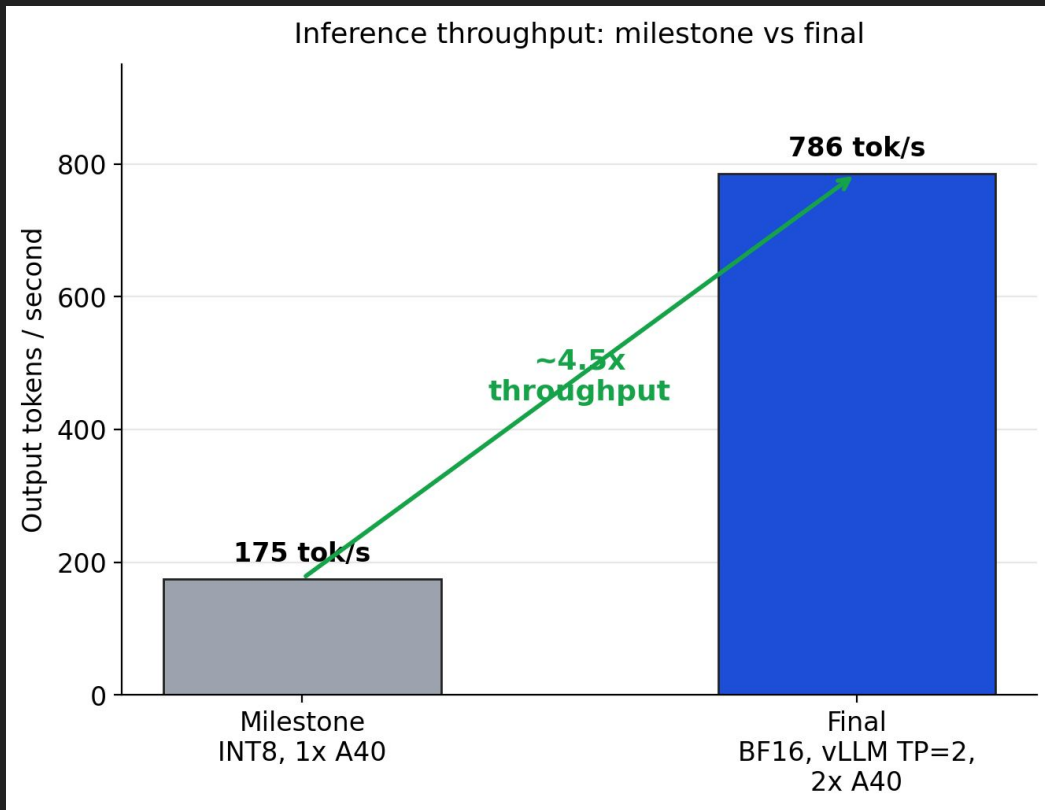
Both setups use <20% of the model's 262,144-token native limit; free-form accuracy stayed ~65% in both.
The wall is reasoning quality, not tokens. (single leaderboard submission each)

Results

Results

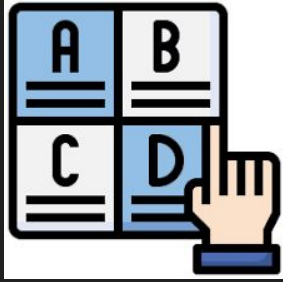


Results

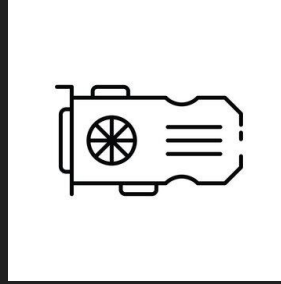


Discussion

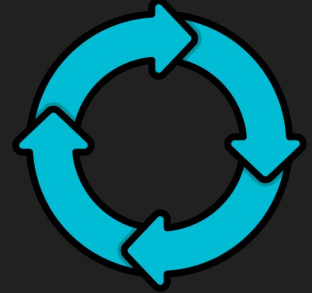
Strength/Weakness



Great result/score on MCQs



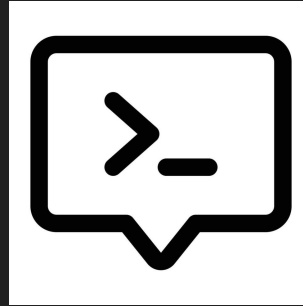
Allocated decent GPU



Continuously running this
for 5 weeks straight



We didn't get to try all of our
ideas due to time constraints



General Prompt wasn't as
optimized



Multiple Answer Score was
weak

Improvements/Learning Moments



Running rapid iterations can allow us to test multiple different experiments at once.

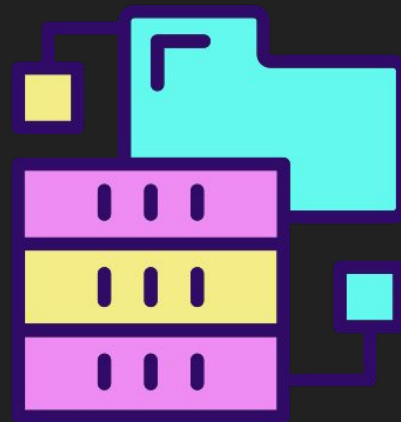


Running the code overnight in a remote server allowed us to run experiments in the background and get back valuable time.

Future Work



Utilize the model context tokens better. Use better prompts and try to answer the multiple select questions better.



Augment the GRPO fine tuning by using NVIDIA's Nemotron dataset as a way to improve post-training model fine tuning.

Thank you!